

WickGrid is a novel, patent-pending geogrid-geotextile composite combining a high-stiffness, open-aperture polymeric geogrid with a unique nonwoven geotextile possessing wicking properties in multiple directions. WickGrid™ is specifically engineered to transport water even under zero hydraulic gradient conditions due to its proprietary microstructure and specialized treatment. WickGrid™ is designed to resist freeze-thaw cycles and mitigate thawing induced differential settlements. WickGrid™ offers moisture management, drainage, reinforcement, confinement, separation, and filtration capabilities in one system.

This high stiffness and integrally formed geogrid is monolithic and isotropic with thick and wide ribs, integral nodes, and uniform square apertures and possesses high stiffness at low strains to be ideal for soil stabilization and base reinforcement. Engineered to be mechanically and chemically stable in aggressive soil environments, WickGrid™ allows sub-base drainage to help maintain structural stability and is highly effective in keeping expensive imported material from being contaminated by migration of fines from saturated base soils.

TESTED PROPERTY	TEST METHOD	UNIT ENGLISH (METRIC)	VALUE ENGLISH (METRIC)	
HIGH STIFFNESS BI-AXIAL GEOGRID			MD	XD
Carbon Black Content	ASTM D 4218	%	2.0	
Radial Stiffness @ 0.5% Strain ⁽¹⁾	ASTM D 6637	lbs/ft (kN/m)	37,697 (550.0)	
Junction Efficiency ⁽²⁾⁽³⁾	GRI-GG2 ASTM D 7737	%	>95	>95
Flexural Rigidity ⁽¹⁾	ASTM D 7748	mg-cm	2,000,000	
Aperture Stability ⁽²⁾⁽⁴⁾	US. COE	kg-cm/deg	11.0	
Minimum Rib Thickness	Callipered	inch (mm)	0.1 (2.5)	0.06 (1.5)
Aperture Size ⁽²⁾⁽⁵⁾	Nominal	inch (mm)	1.50 (38.0)	1.50 (38.0)
NON-WOVEN WICKING GEOTEXTILE				
Grab Strength ⁽¹⁾	ASTM D 4632	lbs (N)	210 (935)	
Trapezoidal Tear ⁽¹⁾	ASTM D 4533	lbs (N)	96 (427)	
CBR Puncture Strength ⁽¹⁾	ASTM D 6241	lbs (kN)	584 (2.6)	
Permittivity ⁽¹⁾	ASTM D 4491	sec ⁻¹	1.52	
Flow Rate ⁽¹⁾	ASTM D 4491	gal/min/ft ² (L/min/m ²)	111 (4,530)	
Horizontal Wet Front Water Movement ⁽⁷⁾ (983 minutes – Zero Gradient)	ASTM C 1599 modified	inch (mm)	120.8 (3,068)	
Maximum Horizontal Wet-Front Velocity ⁽⁹⁾ (Zero Gradient)	ASTM C 1599 modified	in/hr (mm/hr)	196.4 (4,989)	
Water Removal Rate ⁽⁹⁾	See Note	mL/mm/hr	1.113	
Fabric Wettability Rate ⁽⁹⁾	See Note	sec	< 0.033	

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TESTED PROPERTY	TEST METHOD	UNIT ENGLISH (METRIC)	VALUE ENGLISH (METRIC)
Contact Angle ⁽⁹⁾	ASTM D 7334	deg	0
Wicking Sustainability Efficiency (WSE) ⁽¹⁰⁾	Internal Testing Method	%	> 95 > 95
Horizontal			
Vertical			
Apparent Opening Size (AOS)	ASTM D 4751	m	0.075
Mass Per Unit Area ⁽²⁾	ASTM D 5261	oz/yd ² (g/m ²)	6.93 (235)
WICKING GEOGRID COMPOSITE			
Modulus Improvement Factor (MIF) ⁽⁸⁾	AASHTO 1993 modified	-	2.74
Bearing Capacity Improvement Factor ⁽⁸⁾	AASHTO 1993 modified	-	1.58
Hydraulic Improvement Factor (HIF) ⁽⁸⁾	AASHTO 1993 modified	-	1.88
TYPICAL ROLL DIMENSIONS			
Roll Width ⁽⁶⁾		ft (m)	12.95 (3.95) or 19.52 (5.95)
Roll Length ⁽⁶⁾		ft (m)	164.04 (50.0)

NOTES:

(1) Minimum Average Roll Values (MARV) – calculated as mean minus 2X standard deviation as per ASTM -D4759-02.

(2) Average.

(3) Junction efficiency is defined as junction strength divided by multi-rib strength.

(4) Resistance to in plane rotational movement measured at an applied moment = (5.0 kg-cm) in accordance with US Army Corps of Engineers methodology for the measurement of torsional rigidity.

(5) Aperture tolerance: within +/- 10% coefficient of variance. (6) Custom length orders can be accommodated.

(7) Testing conducted at SGI Testing Service, LLC, a GAI-LAP accredited testing facility in the USA. Measurements taken at standard temperature & pressure.

(8) Values determined from results of independent Testing performed at University of Victoria, BC Canada in accordance with the publication titled 'Model tests on wicking geosynthetic composite reinforced bases over weak subgrade.' Geotextiles and Geomembranes Journal, Vol. 53, Issue 4 (August 2025).

(9) Values determined from results of independent Testing performed at University of Victoria, BC Canada in accordance with the publication titled 'Evaluation of water removal capability of wicking nonwoven geotextiles.' Geotextiles and Geomembranes Journal, Vol. 53, Issue 6 (December 2025).

(10) Wicking Sustainability Efficiency is a measurement indicative of long-term sustainability of wicking properties. Wicking Sustainability Efficiency is defined as the wet-front movement at 983 minutes and zero gradient (ASTM C 1599 Modified) of the washed product divided by the wet-front movement at 983 minutes and zero gradient (ASTM C 1599 Modified) of the unwashed product.

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